BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 001 · Data types & unsigned binary

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Choose the most suitable data type for 27 students and for a temperature of 18.5 °C .

Q2. A student ID is "0042" . Which data type would you use, and why would an integer be unsuitable?

Q3. Convert 00101101_2 into denary. Show the place values you add.

Q4. A student says: “The largest unsigned 8-bit value is 256 because $2^8 = 256$.” Explain the mistake.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Recall the term — Integer

Recall the term — Real / floating point

Recall the term — Character

Recall the term — String

Recall the term — Boolean

Recall the term — Bit

Recall the term — Byte

New learning

Choose a type for the purpose — The symbols in a value do not decide its data type on their own. Think about what the value represents and what operations make sense.

Binary → denary — Select only the place values whose bits are 1. — CODE:
00101101_2
= 32 + 8 + 4 + 1
= 45

Denary → binary — Choose place values that add to the target. Write 1 in those columns and 0 in the others. — CODE:
74 = 64 + 8 + 2
$74_{10} = 01001010_2$

Know the range — Eight bits give $2^8 = 256$ different patterns. The range begins at 0, so the largest unsigned value is $2^8 - 1$.

Worked example

Binary → denary

1. Write the columns — CODE:
128 64 32 16 8 4 2 1
0 0 1 0 1 1 0 1

2. Use only the columns containing 1 — CODE:
32 + 8 + 4 + 1

3. Calculate the denary value — CODE:
32 + 8 + 4 + 1 = 45

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

We Do · Practise with guidance

Choose the best type

A school stores this as a student identifier. Which data type is most suitable, and why?

Build the justification

Encode 74

Write 74 as an unsigned 8-bit binary value.

Choose the place values

74 = 64 + 8 + 2.

Check: 64 + 8 + 2 = 74.

Represent an unsigned integer

Convert the denary number 188 to an unsigned 8-bit binary number.

OCR June 2017 question paper .

Build the answer

Why is the maximum 255?

Explain the maximum unsigned value in 8 bits and what happens when we need to store 256.

Complete the rule

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

A weather-monitoring system
For each field, state a suitable data type and justify it:
number of readings collected;
air temperature, e.g. 18.5 °C;
a one-letter category, e.g. A;
station identifier "0042" ;
whether the sensor is working.

Question 2

Encode the readings
Convert each value into unsigned 8-bit binary. Show a place-value decomposition.
CODE:
a) 37
b) 92
c) 200
Classroom adaptation of the conversion skill used in OCR June 2017 Q6(a)(i).

Question 3

Convert unsigned 8-bit binary to denary
Convert each unsigned 8-bit binary number to denary. Show which place values contribute to the total.
CODE:
a) 01101001_2
b) 11000010_2

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Question 4

Choose a storage width
A system must store whole-number readings from 0 to 300 inclusive.
State the largest unsigned 8-bit value.
Explain why 8 bits are insufficient.
Determine the minimum number of bits required and justify it using the unsigned range.

Exit Ticket · Answer before checking

The largest unsigned value that can be represented using 8 bits is 255.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Integer for 27; real / floating point for 18.5. A count uses whole numbers. A temperature may contain a fractional part.

Q2. String . The value is an identifier rather than a number for calculation. A string preserves the leading zeros; an integer would store the numerical value 42.

Q3. 45 32 + 8 + 4 + 1 = 45

Q4. There are 256 different bit patterns , but the unsigned range starts at 0. Eight bits represent 0–255 . The value 256 needs 9 bits: 100000000_2 .

You Do pages · Independent questions

Q1. CODE:
number of readings → Integer
Reason: a whole-number count.

air temperature, e.g. 18.5 → Real / floating point
Reason: may contain a fractional part.

one-letter category → Character
Reason: one character is stored.

station identifier "0042" → String
Reason: it is an identifier and the leading zeros must be preserved.

sensor working → Boolean
Reason: only two states are required: true/false.

Q2. CODE:
37 = 32 + 4 + 1
37 = 00100101

92 = 64 + 16 + 8 + 4
92 = 01011100

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

200 = 128 + 64 + 8
200 = 11001000

Q3. CODE:
01101001
= 64 + 32 + 8 + 1
= 105

11000010
= 128 + 64 + 2
= 194

Q4. CODE:
Largest unsigned 8-bit value:
2^8 - 1 = 255

8 bits are insufficient because:
300 > 255

9 bits give:
0 to 2^9 - 1
= 0 to 511

Minimum required width = 9 bits

Exit Ticket · Model answer

True. Eight bits give 256 values, from 0 to 255.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding